

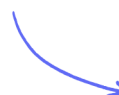
Multiple Choice Questions

Electrons & Ions

Endothermic Ionisation Energy / Electronic Structures / Shapes of Orbitals /
Electronic Configurations & Chemical Properties

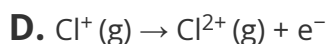
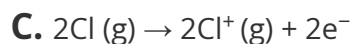
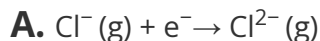
Easy (3 questions)	/3
Medium (4 questions)	/4
Total Marks	/7

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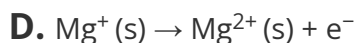
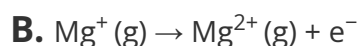
Easy Questions

1 Which equation represents the **second** ionisation energy of chlorine?



(1 mark)

2 Which equation represents the **second** ionisation energy of magnesium?



(1 mark)

3 Several factors may affect ionisation energies:

I) the number of protons increases

II) the outer electron is further from the nucleus

III) the amount of shielding increases

IV) the number of unpaired outer electrons increases

Which factors explain the **decrease** in ionisation energy as Group 1 is descended?

A. I and II

B. II and III

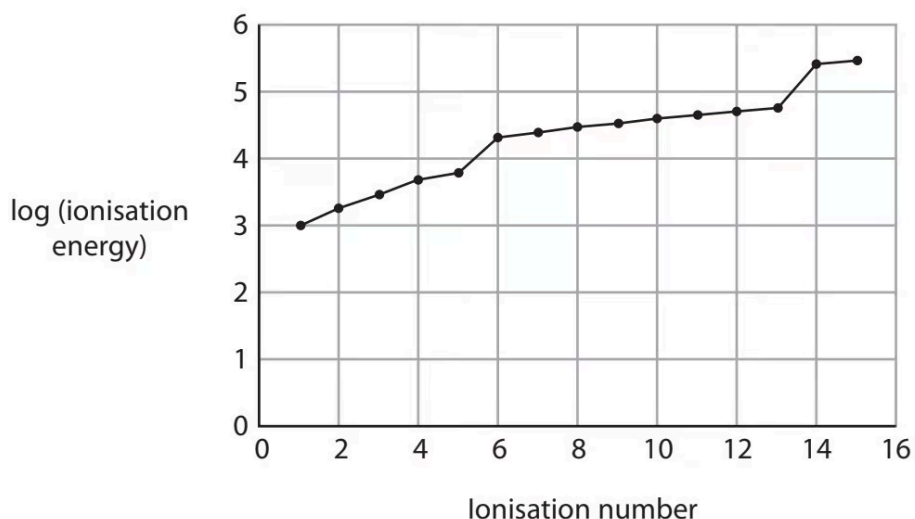
C. III and IV

D. I, II, III and IV

(1 mark)

Medium Questions

- 1 The graph shows $\log$ (ionisation energy) against ionisation number for the successive ionisations of an element.



In this element, how many quantum shells contain electrons, and how many electrons are in the outer quantum shell?

		Number of quantum shells containing electrons	Number of electrons in the outer quantum shell
☐	A	3	2
☐	B	3	5
☐	C	5	2
☐	D	5	5

(1 mark)

2 A stable ion, M^{3+} , contains 18 electrons.

In which block of the Periodic Table is element **M** found?

- A. s
- B. p
- C. d
- D. f

(1 mark)

3 Which ion has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$ in its ground state?

- A. Al^{3+}
- B. Cl^-
- C. N^{3-}
- D. Na^+

(1 mark)

4 A p-block element in **Period 3** of the Periodic Table reacts to form an ionic compound.

What could be the electronic configuration of the **ion** formed by this element?

- A. $1s^2 2s^2 2p^6 3s^2$
- B. $1s^2 2s^2 2p^6 3s^2 3p^6$
- C. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
- D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

(1 mark)